

Eendrag Alumni Hub

Hosting & service costs for 1,000–2,000 alumni · Budget target: R1,500/year

Recommended stack (all free tier)

By moving hosting from Vercel to Cloudflare Pages and staying on free tiers everywhere else, the site runs for the cost of the domain only.

Service	What it does	Plan	Monthly	Annual
Cloudflare Pages	Hosting (SPA + CDN)	Free	R0	R0
Supabase	Database, auth, storage, realtime messaging	Free	R0	R0
Mapbox	Map tiles + geocoding	Free	R0	R0
Domain (.co.za)	e.g. eendragalumni.co.za	Annual reg.	~R15	~R180
		Total	~R15	~R180

Skip the custom domain and use the free eendrag-hub.pages.dev URL for R0/year total.

Free tier limits vs. your usage

Service	Free tier limit	Your usage at 2,000 alumni	Fits?
Supabase MAU	50,000 monthly active users	~2,000	Yes, 25x headroom
Supabase database	500 MB	Text posts / profiles ~5 MB	Yes, 100x headroom
Supabase storage	1 GB (profile photos)	~1 GB uncompressed (tight)	Only with compression
Supabase egress	5 GB / month	Well under	Yes
Mapbox map loads	50,000 / month	Nowhere close	Yes
Mapbox geocoding	100,000 / month	One per city save, cached	Yes
Cloudflare bandwidth	Unlimited	Any	Yes
Cloudflare builds	500 / month	A few per day at most	Yes

What you give up vs. Supabase Pro (\$25/mo)

Concern on free tier	Real impact	Mitigation
Project pauses after 7 days idle	Site goes offline if truly unused	Active alumni site won't sit idle. Optional weekly scheduled ping.
500 MB database limit	Ceiling on text data	You have ~100x headroom. Non-issue unless dumping large JSON
1 GB file storage limit	2,000 raw photos hits the cap	Client-side compress: 400x400 JPEG @ 0.8 = ~50 KB each. Fits 2
No daily backups	Manual recovery in worst case	Weekly manual export from Supabase dashboard, or scheduled ex
Auth email rate limit (~30/hr)	Signup bursts may queue	Add Resend (3K/mo free) only if you actually see this happen.

Cost comparison: current vs. recommended

Path	Hosting	Backend	Monthly	Annual
Current (Vercel + Supabase Free)	Vercel Hobby	Supabase Free	R0	R0*
Vercel Pro + Supabase Pro (safe)	Vercel Pro (\$20)	Supabase Pro (\$25)	~R820	~R9,840
Vercel + Supabase Pro	Vercel Hobby	Supabase Pro (\$25)	~R460	~R5,520
Cloudflare + all free tier	Cloudflare Pages	Supabase Free	~R15	~R180

* Vercel Hobby technically disallows commercial use. Fine for a non-revenue alumni site, but if you ever add donations, merchandise, or paid events you must upgrade to Vercel Pro (~R360/mo) or switch to Cloudflare Pages (which has no such restriction).

What changes when moving Vercel → Cloudflare Pages

Thing	Change
Codebase	Nothing. Same Vite build, same dist/ output.
Environment variables	Re-add VITE_SUPABASE_URL, VITE_SUPABASE_ANON_KEY, VITE_MAPBOX_TOKEN in the
vercel.json (SPA rewrites)	Replace with a public/_redirects file containing: /* /index.html 200
Deploy trigger	Push to GitHub & Cloudflare auto-builds (same as Vercel).
URL	Changes from xxx.vercel.app to xxx.pages.dev. Custom domain unchanged if you have one.
Supabase auth redirect	Update Authentication & URL Configuration & Site URL to the new domain.

Migration steps (approximately 20 minutes)

1. Sign up at pages.cloudflare.com (free, no card required).
2. Connect your GitHub account and select the eendrag-hub repository.
3. Build settings: framework = Vite, build command = npm run build, output directory = dist.
4. Paste the three VITE_ environment variables.
5. Add public/_redirects with the single line: /* /index.html 200 (so React Router works).
6. Deploy and test the eendrag-hub.pages.dev URL.
7. If using a custom domain: add it under Custom domains, update DNS at your registrar.
8. Update Supabase Site URL to the new domain so auth confirmation emails point correctly.
9. Optionally delete the Vercel project (or keep it as a warm backup).

The one code change needed: image compression

The Supabase free tier's 1 GB storage limit is the only real risk. 2,000 users at ~500 KB per uncompressed avatar = 1 GB exactly. Fix by resizing photos client-side before upload (400x400, JPEG quality 0.8) which brings each photo down to ~50 KB. That gives roughly 20,000 photos of headroom instead of 2,000. Same technique applies to event, business and post images. This is roughly a 20-line change to the upload paths.

Bottom line

Move Vercel → Cloudflare Pages, stay on Supabase Free, add client-side image compression. Total cost: **~R180/year for the domain**, comfortably inside a R1,500/year budget with roughly R1,320/year of headroom

for a future Supabase Pro upgrade or a scheduled backup service if needed.